

BOOK REVIEW

Wolfram Saenger, *Principles of Nucleic Acid Structure*, Advanced Texts in Chemistry Series, Springer-Verlag, New York, 1983, 556 pp., \$44.00

The goal of the Advanced Texts in Chemistry Series, as indicated by the series editor, C. R. Cantor, is to give a relatively concise but instructive introduction to specific areas of chemistry in which recent progress has outpaced what is covered in any available textbooks.

The phrase "nucleic acid structure" has often been used to indicate base sequence (primary structure) or base pairing (secondary structure) in the literature. However, as used in this book, the word "structure" in most cases refers to the three-dimensional arrangement of individual atoms within the molecule. Primary and secondary structures are only discussed where they can either help to clarify or to give a better understanding of the three-dimensional structure.

Starting from the question "Why study nucleic acid structure?" the author indicates that his main purpose is to understand the function (both biological and chemical) of DNA and RNA. Throughout the text, whenever possible, the functional aspects are mentioned to illustrate the structural features of the molecule.

Nucleic acid structure is a rapidly growing field. The left-handed helix was not accepted as a structure of DNA in the early 1970s; now several Z-DNAs have been crystalized. In addition to the now widely accepted structures (e.g., A-, B-, C-, D-, Z-DNA, and A-RNA), some less accepted or postulated structures (e.g., side-by-side model, vertical double helix) of nucleic acids are discussed.

Overall, this book is very well organized and well written. It is an excellent textbook with relatively new information included (e.g., Calladine's rule to predict the sequence-dependent local variations of DNA double helix). The completeness plus the up-to-date information make this book a good tool for theoretical biologists who want to study the structure and function of nucleic acid. An outstanding list of references (over 1,300) makes it also a useful handbook for new researchers who want to enter this field.

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